

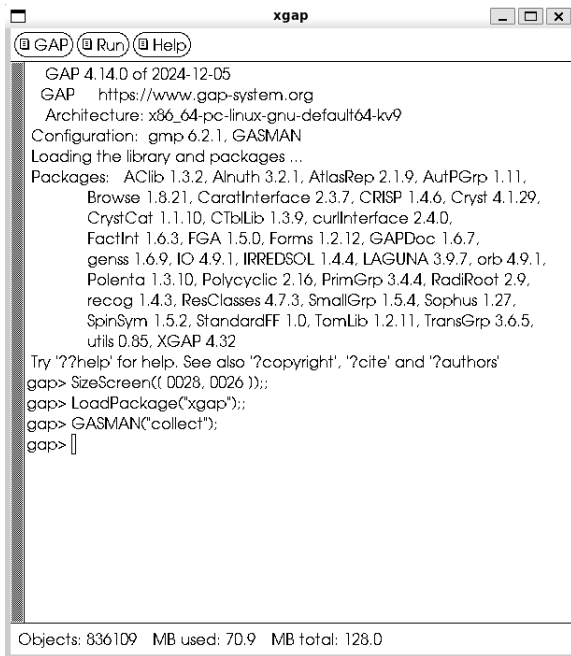
Visualising GAP objects: current status and the future

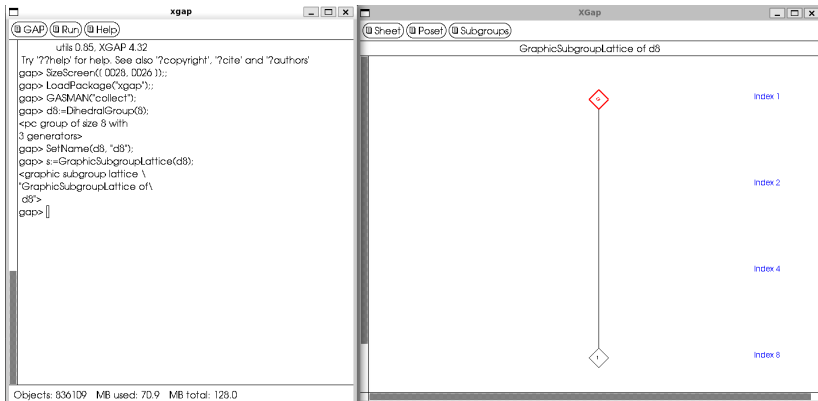
Lukas Schnelle

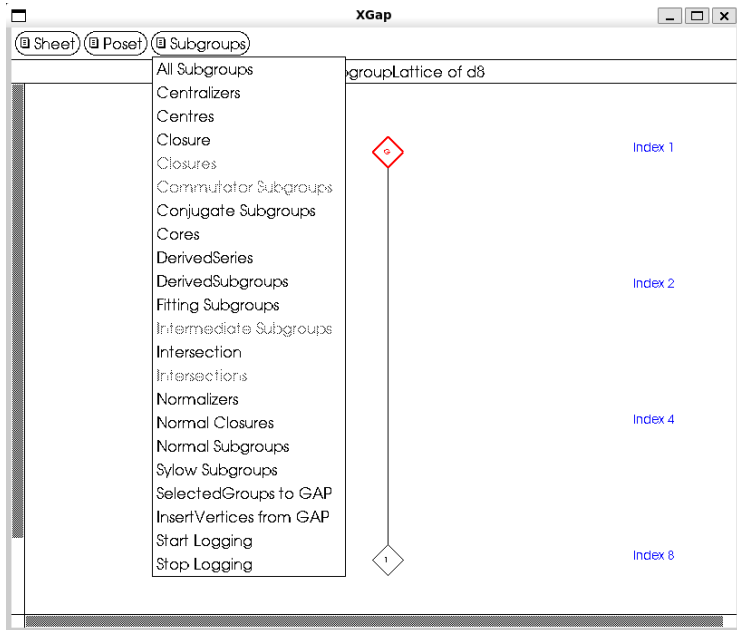
GAPDays Summer 2026

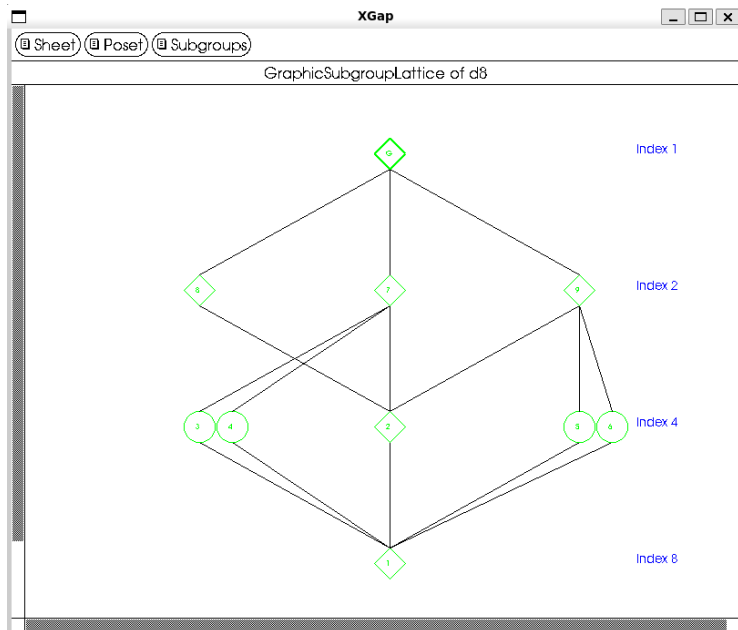
TODO: motivation

Currently there are multiple packages, which provide functionality to visualise GAP objects. First we will give an (possibly incomplete) overview of them with some examples.









Advantages

- Direct GUI to use
- Freely movable visualisation

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Challenges

- Needs to be started standalone
- Needs to be adjusted to every OS

francy (by Manuel Martins)

Idea

Allow the XGAP system to run in a Jupyter Notebook.

Status

I was not able to get this running.

```

LoadPackage("digraphs"); LoadPackage("francy");

G := SymmetricGroup(3); as := AllSubgroups(G); nodes := [];
d := Digraph(as, {H, K} -> IsSubgroup(H, K));

vertices := DigraphVertices(d); edges := DigraphEdges(d);

canvas := Canvas(Concatenation("Subgroups Digraph of ",
    String(G)));
graph := Graph(GraphType.DIRECTED);
Add(canvas, graph);

customMessage := FrancyMessage(FrancyMessageType.INFO,
    "Simple Groups", "A group is simple if it is nontrivial
    and has no nontrivial normal subgroups.");

IsGroupSimple := function(i)
    Add(canvas, simpleGroupMessage);
    if IsSimpleGroup(as[i]) then
        Add(canvas, FrancyMessage("Simple",
            Concatenation("The vertex ", String(i),
                ", representing the subgroup ", String(as[i]),
                ", is simple.")));
    else
        Add(canvas, FrancyMessage("Not Simple",
            Concatenation("The vertex ", String(i),
                ", representing the subgroup ", String(as[i]),
                ", is not simple.")));
    fi;
    return Draw(canvas);
end;

```

```
for i in vertices do
  nodes[i] := Shape(ShapeType.CIRCLE, String(i));
  Add(nodes[i], Menu("Is this subgroup simple?",
    Callback(IsGroupSimple, [i])));
  Add(graph, nodes[i]);
od;

for i in edges do
  Add(graph, Link(nodes[i[1]], nodes[i[2]]));
od;

Draw(canvas);
```

(from [1])

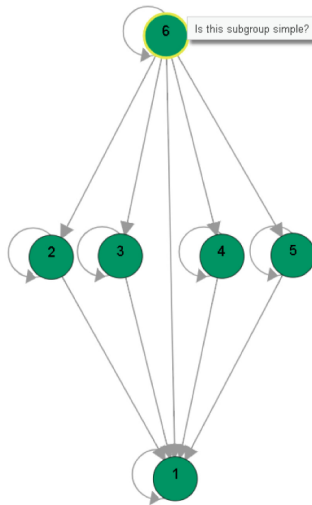
Francy

Subgroups Digraph of $\text{SymmetricGroup}([1..3])$

Simple Groups A group is simple if it is nontrivial and has no nontrivial normal subgroups.

Simple The vertex 2, representing the subgroup $\text{Group}([2,3])$, is simple.

Not Simple The vertex 6, representing the subgroup $\text{SymmetricGroup}([1..3])$, is not simple.



(from [1])

Advantages

- Runs in browser
- Highly customisable

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Other packages

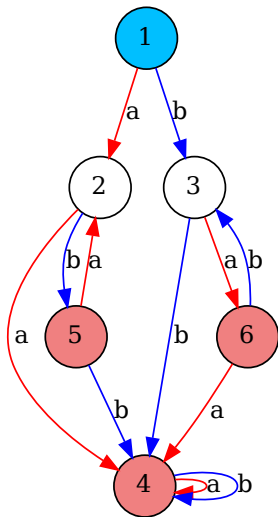
- jupyterviz
- semigroupviz

SgpViz (by Manuel Delgado and José João Morais)

Semigroup visualisation

- Can visualise e.g. Cayley graphs
- Using graphviz

```
gap> f := FreeMonoid("a","b");;
gap> a := GeneratorsOfMonoid( f )[ 1 ];;
gap> b := GeneratorsOfMonoid( f )[ 2 ];;
gap> r:=[a^3,a^2], [a^2*b,a^2], [b*a^2,a^2], [b^2,a^2], [a*b*a,a], [b*a*b,b] ;;
gap> b21:= f/r;
<fp monoid on the generators [ a, b ]>
gap> DrawCayleyGraph(b21);
```



Advantages

- Uses graphviz standard
- Outputs PDFs

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Challenges

- Needs additional installation of graphviz on system

Automata (by Manuel Delgado, Steve Linton and José João Morais)

Automata visualisation

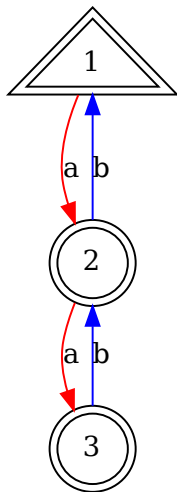
- Using graphviz

Automata (by Manuel Delgado, Steve Linton and José João Morais)

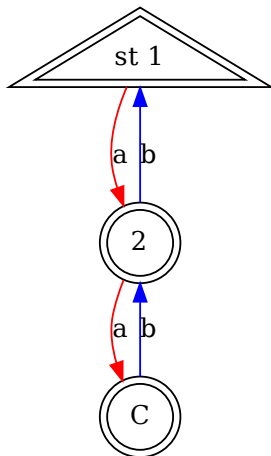
Automata visualisation

■ Using graphviz

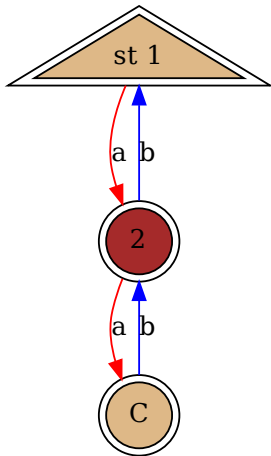
```
gap> x:=Automaton("det",3,2,[ [ 2, 3, 0 ], [ 0, 1, 2 ] ],[ 1 ],[ 1, 2, 3 ]);;
gap> DrawAutomaton(x);
```




```
gap> DrawAutomaton(x, ["st 1", "2", "C"]);
```



```
gap> DrawAutomaton(x, ["st 1", "2", "C"], [[2], [1, 3]]);
```



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graphviz (by James D. Mitchell and Matthew Pancer)

Digraphs visualisation

- (soon) extension to the Digraphs package
- Also uses graphviz standard

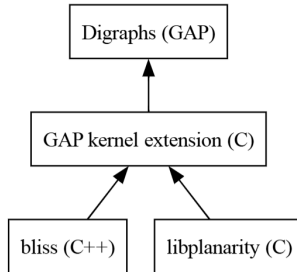
graphviz (by James D. Mitchell and Matthew Pancer)

Digraphs visualisation

- (soon) extension to the Digraphs package
- Also uses graphviz standard

```
gap> D := Digraph([], [1], [2], [2]);  
<immutable digraph with 4 vertices, 3 edges>  
gap> SetDigraphVertexLabels(D,  
> ["Digraphs (GAP)",  
> "GAP kernel extension (C)",  
> "bliss (C++)",  
> "libplanarity (C)"]);  
gap> gv := GraphvizVertexLabelledDigraph(D);  
<graphviz digraph "hgn" with 4 nodes and 3 edges>  
gap> GraphvizSetAttr(gv, "node [shape=box]");  
<graphviz digraph "hgn" with 4 nodes and 3 edges>  
gap> GraphvizSetAttr(gv, "rankdir", "BT");  
<graphviz digraph "hgn" with 4 nodes and 3 edges>
```

```
gap> Print(AsString(gv), "\n");
//dot
digraph hgn {
  node [shape=box] rankdir=BT
  1 [label="Digraphs (GAP)"]
  2 [label="GAP kernel extension (C)"]
  3 [label="bliss (C++)"]
  4 [label="libplanarity (C)"]
  2 -> 1
  3 -> 2
  4 -> 2
}
```



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- Uses graphviz standard
- Is a package extension
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- Visualisations can be iterated

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Challenges

- Needs GAP function for everything in graphviz standard
- No immediate way to show results without additional installation on system

```
gap> Splash(gv);  
Error: /tmp/gaptempdirv6HuGo/hgn.dot: syntax error in line 4 near '('
```

IntPic (by Manuel Delgado)

Visualises arrays of integers

Can be used to highlight different properties in large number of colours.

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Can be used to highlight different properties in large number of colours.

```
gap> rg := [-5..23];;  
gap> len := 10;;  
gap> hg := rec();;  
gap> hg.highlights:=[[2,3,5,7],[11,13,17,19],[23]];;  
gap> tkz := IP_TikzArrayOfIntegers(rg,len,hg);;  
gap> Print(tkz);
```

```
gap> Print(tkz);
%tikz
\begin{tikzpicture}[every node/.style={draw,scale=1pt,
minimum width=20pt,inner sep=3pt,
line width=0pt,draw=black}]
\matrix[row sep=2pt,column sep=2pt]
{\node[]{}{15};&
\node[]{}{16};&
\node[fill=green]{}{17};&
\node[]{}{18};&
\node[fill=green]{}{19};&
\node[]{}{20};&
\node[]{}{21};&
\node[]{}{22};&
\node[fill=blue]{}{23};\\
\node[fill=red]{}{5};&
\node[]{}{6};&
\node[fill=red]{}{7};&
\node[]{}{8};&
\node[]{}{9};&
\node[]{}{10};&
\node[fill=green]{}{11};&
\node[]{}{12};&
\node[fill=green]{}{13};&
\node[]{}{14};\\
\node[]{}{-5};&
\node[]{}{-4};&
\node[]{}{-3};&
\node[]{}{-2};&
\node[]{}{-1};&
\node[]{}{0};&
\node[]{}{1};&
\node[fill=red]{}{2};&
\node[fill=red]{}{3};&
\node[]{}{4};\\
};
\end{tikzpicture}
```

15	16	17	18	19	20	21	22	23	
5	6	7	8	9	10	11	12	13	14
-5	-4	-3	-2	-1	0	1	2	3	4

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- No immediate way to show results without compiling them

GAPic (by Alice C. Niemyer and many more)

Visualising triangulated polyhedra

- Came from the `SimplicialSurfaces` package
- Expanded functionality like general polyhedra
- Has `SimplicialSurfaces` package as extension

```
gap> oct := Octahedron();;
gap> verticesPositions := [
> [ 0, 0, Sqrt(2.) ],
> [ 1, 1, 0 ],
> [ 1, -1, 0 ],
> [ -1, -1, 0 ],
> [ -1, 1, 0 ],
> [ 0, 0, -Sqrt(2.) ] ];;
gap> printRecord := SetVertexCoordinates3D(oct, verticesPositions, rec());;
gap> DrawComplexToJavaScript(oct, "octahedron.html", printRecord);;
```

Demonstration in browser


```

gap> oct := Octahedron();;
gap> verticesPositions := [
> [ 0, 0, Sqrt(2.) ],
> [ 1, 1, 0 ],
> [ 1, -1, 0 ],
> [ -1, -1, 0 ],
> [ -1, 1, 0 ],
> [ 0, 0, -Sqrt(2.) ] ];;
gap> printRecord := SetVertexCoordinates3D(oct, verticesPositions, rec());;
gap> DrawComplexToJavaScript(oct, "octahedron.html", printRecord);;

```

Demonstration in browser

```

gap> printRecord := SetFaceColour(oct, 3, "0x00FF00", printRecord);
rec( edgeThickness := 0.03, faceColours := [ , "0x00FF00" ],
  vertexCoordinates3D := [ [ 0, 0, 1.41421 ], [ 1, 1, 0 ], [ 1, -1, 0 ], [ -1, -1, 0 ], [ -1, 1, 0 ],
    [ 0, 0, -1.41421 ] ] )
gap> DrawComplexToJavaScript(oct, "octahedron-coloured.html", printRecord);;

```

Advantages

- Results can be opened with any browser
- High interactivity
- Can use parameterised coordinates

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- High interactivity
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Challenges

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There are certainly more packages that include some visualisation which I did not mention.

Goals

Make visualisations in GAP...

- more visible
- more intuitive → accessible
- sustainable

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Make visualisations in GAP...

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Methods

- Use of package extensions?
- Use of existing standards

- What are your experiences with visualisations in GAP?
- What do you expect from a visualisation functionality?
- Is there anything you want to visualise but are not able to visualise yet?